

# What an LLM-judged evaluation of ambient clinical scribes can and cannot see: an open harness and two case studies

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Draft, September 2026

## 0.1 Abstract ( $\approx$ 200 words)

Ambient clinical scribes turn a recorded consultation into a draft note, and are increasingly evaluated on clinician-defined dimensions scored by LLM judges. We release an open harness that reproduces that style of evaluation on public data (Pri-Mock57 audio, ACI-Bench notes, a 7,092-term primary-care lexicon) and use it for two studies. Study 1 evaluates the after-visit pipeline. Across seven open speech recognizers, word error rate varies within a point while medical-term loss varies twofold, separating language-model decoders (8.4% [7.1, 9.7]) from classic decoders (12.5–15.1%); the effect replicates on a synthetic second corpus. Speaker diarization alone does not change the note; naming the clinician before writing cuts misattributions by two thirds under one judge, a result three judges then fail to agree on. An overnight search over 131 pipeline variants gains term precision (+4.5 [+2.3, +6.7]) but its composite gain (+4.2 [−0.3, +8.8]) is not resolved at  $n = 37$ , and the search removed span citations because its objective had no verifiability term; re-scored with one, citations halve unsupported sentences (+5.3 grounded [+3.2, +7.8]). A quarter of reference plan items were never spoken, exposing a ground-truth flaw in completeness metrics. Study 2 evaluates an in-visit decision-support view on the same visits: prompt rules reduce red flags from 61 to 3 (all judged genuine) without losing diagnosis coverage (56/57), while guideline retrieval adds volume, not agreement. Three judges agree on ranking ( $\rho = 0.80$  between the two scoring the composite) but not on attribution: pairwise  $\kappa$  from  $-0.02$  to  $+0.29$ , no note flagged by all three, and the third judge reverses the direction of the attribution difference between pipelines; a reinforcement-learning pass the first two disagree on in sign. We argue these evaluations are useful for ordering pipelines and for catching objective failures, and unsafe as absolute quality claims without clinician calibration.

## 0.2 1. Introduction

- Ambient scribes are deployed at scale, and the evaluation method described in the vendor and research literature is clinician-defined dimensions (attribution, completeness, fairness) scored by LLM judges validated against expert review.

- Independent, reproducible versions of that evaluation do not exist on open data. We built one, ran it end to end on open models and open audio, and report what it found and what it could not see.
- Contributions:
  1. **scribe-bench**: an open harness over PriMock57 and ACI-Bench with seven ASR runners, note generation against any OpenAI-compatible server, a claim-level verifier with an injected-error benchmark, LLM-judge scorers for attribution, completeness and grounding, an autoresearch loop with a held-out gate and a second judge, and a replayed in-visit decision-support view with its own timeline scoring. Code, lexicon, splits, prompts and cached outputs released.
  2. **Findings that transferred** across a change of data, judge or metric: decoder type drives medical-term loss; role mapping, not diarization, fixes attribution; span citations trade completeness for grounding; overlap metrics and clinician dimensions diverge under fine-tuning.
  3. **Findings about the evaluation itself**: an objective without a verifiability term removes the verifiability feature; 27% of reference plan items are unspoken, so completeness against the note over-credits guessing; capable judges rank pipelines alike ( $\rho = 0.80$ ) but three of them disagree on attribution (pairwise  $\kappa -0.02$  to  $+0.29$ , no note flagged by all three) and two disagree on the sign of an RL result; a model judging its own outputs cannot detect its own escalation errors.
  4. **An in-visit case study** showing that prompt-level rules, not retrieval, moved the safety numbers, with confidence intervals.

### 0.3 2. Setup

**Data.** PriMock57: 57 mock UK GP consultations with separate doctor and patient audio tracks (mixed to mono), human transcripts with per-speaker timestamps, and the clinician’s note; split 20 dev / 37 test by sorted id. ACI-Bench: 120 encounters (three test splits) with human, ASR and corrected transcripts and reference notes; used for note generation, fine-tuning evaluation, the verifier benchmark and fairness stratification. Lexicon: 7,092 primary-care terms from MTS-Dialog and MedSynth; 1,741 term occurrences in the PriMock human transcripts. Synthetic second audio corpus: ACI-Bench dialogues rendered with a two-voice TTS.

**Pipelines.** Vanilla: streaming ASR (Nemotron-3.5 0.6B, 1.1 s chunks, no speaker labels) → untuned Qwen3.8-27B note. Two-pass: after-visit ASR with an LLM decoder and speaker labels (MOSS-Transcribe-Diarize 0.9B; NVIDIA equivalents Canary-Qwen 2.5B + Sortformer) → optional role mapping (S01/S02 → Doctor/Patient by a 60-token LLM call) → note model → optional span citations → verifier. Search knobs: ASR variant, transcript correction, role map, prompt, extra instruction, citations, verifier gate, proposition scaffold, note model (Qwen3.8-27B local, DeepSeek V4 Flash).

**Metrics.** Transcript: WER after symmetric normalization; medical-term miss rate; DER (250 ms collar); word-level speaker misattribution. Note: lexicon term recall/precision; ROUGE-L; plan-item recall (judge extracts items from the reference note, checks each in the draft; variant: transcript-stated items only); misattributions

per note (judge, four types); grounded (judge, whole human transcript as evidence); cited fraction; linked (judge, only the cited lines as evidence); edit effort. Composite for the search:  $0.3 \text{ TR} + 0.3 \text{ TP} + 0.2 \text{ ROUGE-L} + 0.2 \text{ plan} - 40 \times \text{misattrib}$ . Verifier: recall on injected errors (negation, number, drug swap, fabricated finding, third-party leak) over 2,764 claims.

**Judges.** Judge 1 Qwen3.8-27B-FP8 (vLLM, thinking off) for all per-note judgments; judge 2 DeepSeek V4 Flash for confirmation of accepted configs. Judge benchmark on injected errors: Qwen 99.1% recall / 6.5% false positive, DeepSeek 95.7% / 4.5%, Gemma 4 98.7% / 5.5%.

**Statistics.** Paired bootstrap over visits (10,000 resamples) for differences; binomial or Wilson intervals for rates; Spearman  $\rho$  and Cohen’s  $\kappa$  for judge agreement. All reported in runs/paper\_stats.md.

**Compute.** One workstation with two RTX PRO 6000 (Blackwell) GPUs and a Mac Studio; about \$49 of rented H200 time for fine-tuning. Everything else local.

## 0.4 2a. Related work (every reference below verified against the arXiv, ACL Anthology, PMC, Europe PMC or medRxiv record on 13 Sep 2026)

- **Datasets.** PriMock57 [1]: 57 mock primary-care consultations with audio, utterance-level transcripts and notes, released as an ASR and note-generation benchmark; we add per-speaker timing exports and a dev/test split. ACI-Bench [2]: 207 role-played encounters with human, ASR and corrected transcripts and notes across three test sets; our note-generation, fine-tuning, verifier and fairness studies use its test splits. MedMosaic [3] is a large medical-audio question-answering benchmark (46,701 pairs) for multimodal models; the Multicultural Medical Assistant study [4] asks whether LLMs can correct medical ASR errors across accents. Our medical-term miss rate is a narrower, lexicon-based view of the same problem.
- **Evaluation practice in deployed systems.** Published vendor methodology describes clinician-defined dimensions scored by LLM judges validated against expert review, hallucination-detection frameworks, and clinician-in-the-loop studies before deployment. Our attribution, completeness and grounding judges reproduce that shape on open data; our verifier’s injected-error recall (98.7–99.6%) is an open analogue of a hallucination-detection rate, with the caveat that ours is measured on synthetic corruptions.
- **LLM-as-a-judge in healthcare.** Two 2026 scoping reviews catalogue the field: Li et al. [5] analyse 134 studies from January 2023 to February 2026 with a focus on human alignment, and the MedJUDGE review [6] covers 49 studies from 2020 to January 2026 and proposes a governance framework. Reported alignment ranges widely: on multi-document EHR summaries an LLM judge reached an intraclass correlation of 0.82 with clinicians on a validated instrument [7]; on physician-to-physician eConsult concordance the best judge reached  $\kappa = 0.75$  against a physician majority vote while the three physicians agreed among themselves only moderately (Fleiss  $\kappa = 0.46$ ) [8]. DeLucia et al. [9] show judges and clinicians reaching the same completeness verdicts for different reasons; Sun

et al. [10] contrast similarity metrics with LLM judges on clinical dialogue. Our contribution to this line is narrower and mechanical: capable judges agree on ranking ( $\rho = 0.80$ ) and disagree on a low-base-rate dimension (attribution, pairwise  $\kappa -0.02$  to  $+0.29$  across three judges), and an objective built from judged terms can remove a feature the judges never see.

- **Rubric-based clinical evaluation.** HealthBench [11] uses rubrics written by 262 physicians (48,562 criteria); DistillNote [12] evaluates note summaries by their usefulness for downstream clinical prediction. Our plan-item audit (27% of reference items unspoken) is an instance of the ground-truth problem such rubrics also face when the reference note contains decisions that were never verbalized.
- **Medical ASR beyond WER.** Afonja et al. [13] show that low overall WER coexists with high error on clinical entities and that fine-tuning on accented clinical speech closes much of the gap; Adedeji et al. [4, 14] use LLMs to repair medical transcription errors; Ellis et al. [15] show WER and a suite of standard metrics correlate poorly with clinician-assigned risk labels and train a judge to replicate the clinical assessment; MedWER [16] proposes a model-free term-list protocol. Our lexicon-based term-miss rate sits between WER and those judged-impact scores: model-free like MedWER, but reported with the decoder as the explanatory variable.
- **Diarization.** Sortformer [17] is the end-to-end diarizer we pair with Parakeet; the finding that anonymous speaker labels do not reach the note until roles are assigned is, to our knowledge, not reported elsewhere.
- **Note generation models.** MEDIQA-Chat 2023 systems [18] established GPT-4 in-context generation as competitive with human notes on ACI-Bench; Wang et al. [19] adapt an open 13B model with continued pretraining, SFT and RL from AI and human feedback; Ahmed and Sadeque [20] compare parameter-efficient fine-tunes on MTS-Dialog and report persistent hallucination. Our fine-tuning result (overlap up, clinician dimensions down, every interval excluding zero) is the same tension measured on both sides at once.
- **Judge bias and reward hacking.** Wataoka et al. [21] trace self-preference in LLM judges to a preference for low-perplexity text; Ye et al. [22] quantify twelve judge biases; Norman et al. [23] show exact-match agreement overstates judge discrimination and that  $\kappa$  deflates agreement by 33–41 points, which matches our attribution result. Khalaf et al. [24] show the rise-then-fall pattern of inference-time reward hacking is inevitable under common alignment methods; Nishimura-Gasparian et al. [25] show RL reasoning training increases specification gaming. Our search loop removing citations, and our expert-iteration pass learning style but not content, are small instances of the same dynamics under an LLM-judged objective.
- **Ambient documentation systems.** Commercial scribes publish latency and adoption figures but not reproducible evaluations; open-weight speech and note models (NVIDIA Nemotron/Parakeet/Canary, MOSS-Transcribe-Diarize, Qwen, DeepSeek) make an independent evaluation feasible, which is what this paper does.

### 0.4.1 References (verified)

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Corrections made during verification: the earlier draft’s “7 studies in 2024 to 81 in 2025” growth figure was a search-snippet artefact and is not in either review; the  $\kappa = 0.75$  and Fleiss  $\kappa = 0.46$  figures belong to the eConsult preprint [8], not to the npj paper [7], whose result is an ICC of 0.82.

## **0.5 3. Study 1: the after-visit note**

### **0.5.1 3.1 The decoder, not latency, decides medical-term loss**

Table 1 (seven systems, 57 recordings). WER spans 10.3–11.8%; medical-term miss spans 8.4–15.1%. Language-model decoders (MOSS-TD 8.4% [7.1, 9.7]; Canary-Qwen 9.4% [8.0, 10.8]) versus transducer decoders (Nemotron streaming 12.5% [10.9, 14.0]; Parakeet-TDT v3 15.1% [13.4, 16.8]); intervals do not overlap. Running the streaming model offline on the whole recording changes nothing (12.5 vs 12.7). With the recording as the sampling unit (paired bootstrap over 57), MOSS-TD minus streaming is  $-3.9$  points of term miss [ $-6.3, -1.7$ ] and  $-1.4$  WER [ $-1.8, -1.0$ ]; Canary-Qwen minus streaming is  $-2.6$  term miss [ $-4.8, -0.6$ ] at unchanged WER ( $-0.1$  [ $-0.4, +0.2$ ]), so the decoder effect is separable from overall accuracy. Complaint-tuned hotwords make WER measurably worse ( $+0.18$  [ $+0.03, +0.33$ ]) and do not change term miss. Replicates on the synthetic ACI corpus (MOSS 6.6% vs Nemotron 8.9%). Diarization: MOSS-TD DER 11.4%, Sortformer + Parakeet 11.1%,  $\approx 1\%$  of words misattributed for both; Nemotron 3 Nano Omni prompted as a transcriber was unusable (27% WER, DER 88%).

### **0.5.2 3.2 Diarization alone does not reach the note; role mapping does**

Notes from the diarized transcript with anonymous labels carried the same misattribution rate as notes from the undiarized streaming transcript (0.19 per note on all 57); notes from the human transcript, 0.05. On the held-out set the vanilla pipeline and the anonymous-label two-pass pipeline tie on every note metric (composite  $-1.0$  [ $-5.9, +4.1$ ]). Mapping S01/S02 to Doctor/Patient before writing was the first change the search accepted (dev 44.3  $\rightarrow$  45.7, test 39.6  $\rightarrow$  41.8) and, in the final configuration, misattributions are 0.03 per note versus 0.08–0.11.

### **0.5.3 3.3 What an overnight search found, with intervals**

131 iterations; gate = dev  $\geq$  best  $- 0.1 \rightarrow$  held-out test must beat best  $\rightarrow$  judge 2 within 0.5. Accepted: role map ( $+0.9$  dev), DeepSeek writer ( $+1.6$ ), complaint-tuned hotwords + a precision-first instruction ( $+0.8$ ). Held-out composite: vanilla 40.8, best 45.0; difference  $+4.2$  [ $-0.3, +8.8$ ] under judge 1 and  $+2.5$  [ $-0.1, +5.3$ ] under judge 2. Components that are resolved: term precision  $+4.5$  [ $+2.3, +6.7$ ], ROUGE-L  $+2.2$  [ $+1.0, +3.2$ ]. Not resolved: term recall, misattribution, plan recall, grounding. Every proposer-written prompt variant after the acceptance, transcript correction (six times), the scaffold and the verifier gate gained on dev and failed on test. The search is a benchmark-maximizer by construction; the gate is the useful part.

### **0.5.4 3.4 The objective removed citations**

Citations cost  $\sim 1$  ROUGE-L and buy 4–5 term-precision points on every ACI split, but the composite has no verifiability term, so the search rejected citations eight times (the recall cost is all it could see). Re-scored with grounded/linked: cited  $-$

best is +5.3 grounded [+3.2, +7.8] and +2.0 precision [−0.5, +4.6] against −1.2 ROUGE-L [−2.3, −0.2] and −6.5 plan recall [−13.3, +0.2]; 88.8% of sentences cited, 84.0% supported by their own cited lines; unsupported sentences 11.4% → 6.8%. The uncited best is no better grounded than vanilla (−0.8 [−4.0, +2.4]). Lesson stated as such: a score without the trust dimension removes the trust feature.

### **0.5.5 3.5 Completeness against the note over-credits guessing**

Item-level audit of the held-out plan items: 41 of 154 reference plan items were never stated in the conversation (silent test orders, drugs written but not discussed). On spoken items the cited writer captures 101/113 vs 105/113 uncited vs 110/114 vanilla; on unspoken items 16 vs 19 vs 11. Half the cited writer’s headline plan-recall gap is refusal to invent. Four prompt variants aimed at plan recall gained up to +1.3 on dev and none held on test. Recommendation: score completeness against transcript-stated items, which is also closer to a completeness-of-follow-ups dimension as used in clinician-defined rubrics.

### **0.5.6 3.6 Overlap metrics and clinician dimensions diverge under fine-tuning (ACI-Bench)**

One epoch of LoRA on 2,804 open pairs, per-encounter paired bootstrap over 120 ACI-Bench encounters (Table 5): ROUGE-L +8.6 [+7.2, +10.0], term recall +4.8 [+2.9, +6.6], term precision +10.2 [+7.9, +12.6]; and, judged by the same model, misattributions per note +0.11 [+0.04, +0.18] and follow-up recall −4.5 [−7.3, −2.0] (97.4 → 92.9). Every interval excludes zero: the tune that wins every overlap metric loses on both clinician dimensions. Citation behaviour was erased (99.6% → 9–16% cited) because the targets carried none. Untuned Nemotron 3 Ultra 550B sits a point below untuned Qwen 27B on ROUGE-L with the fewest misattributions of any base model. Ordering that held: tuning > base quality > size.

### **0.5.7 3.7 A reinforcement-learning pass the judges disagree about**

Expert iteration on the local writer (sample 8, keep 2, LoRA, four rounds, reward = composite + grounding + linking with a length band; predictions logged before the first sample). Dev 46.2 → 51.0. Held-out: 41.5 → 40.5 under judge 1, 44.3 → 45.9 under judge 2. Style transferred (99.5% cited, notes 19% shorter, edit effort −25%); content did not (term recall −2). None of the predicted reward-hacking signatures appeared in four rounds. At 20 training transcripts this is memorization, and the sign disagreement between judges is the point.

## **0.6 4. Study 2: in-visit decision support on the live transcript**

**Setup.** The streaming transcript is replayed on the human utterance timeline; every 20 s of visit time the model (DeepSeek V4 Flash, 220 tok/s) updates a JSON view: differential with evidence and what is missing, the single discriminating question, red

flags (tiered), plan stated vs suggested, safety-netting given vs to add, assumptions under test, announced revisions. Scoring against the clinician’s note and the timed transcript: time to the reference diagnosis in the top-3 and lead over the moment the GP stated it; suggested questions later asked; red-flag precision; plan suggestions agree/extra/contradict; revisions toward/away; premature complaint; stability; latency. 57 visits, 1,566 ticks per version.

**Versions.** v2 first prompt; v4 danger-first + consult modality + simple causes; v5 five-slot differential with a time-critical tag, flags only for stated features with a tier, a 90 s hold enforced in code; v5+NHS with guideline pages retrieved per differential entry (NHS A-Z → patient.info → Wikipedia low-trust; 191 lookups).

**Results (Table 3).** Diagnosis ever in top-3: v2 52/57 [81, 96], v5 56/57 [91, 100]; v5 – v2 +0.07 [+0.02, +0.14]. Lead over the GP: median 5:36, in 54/54 visits where the GP stated a diagnosis. Red flags: v2 61 raised / 7 false; v5 3 raised / 0 false; per-visit –1.02 [–1.44, –0.63] flags and –0.12 [–0.23, –0.04] false alarms. Contradicting plan suggestions: 13 → 3, –0.18 [–0.33, –0.04] per visit. v4’s danger-first rule raised agreeing suggestions (+0.79 [+0.39, +1.21] per visit) but evicted the likely diagnosis (top-3 –0.05 [–0.12, +0.02]) and produced hedged flags. Guideline retrieval (v5+NHS – v5): no resolved change on any measure; suggestions +31 mostly “extra”; the NHS anaphylaxis page re-imported emergency advice on an insect-sting visit the prompt rules had calmed.

**Reading.** The safety improvements are prompt rules (stated-feature-only flags, a hold, tiering), and they are resolved at  $n = 57$ . “Ahead of the GP” measures how fast the view reads the history, not diagnostic insight. Every flag and contradiction is a model judging a model; the three v5 flags and three contradictions are what a clinician should read first.

## 0.7 5. What the judges can and cannot see

- **Ranking, yes.** Per-note composites from the two judges correlate at  $\rho = 0.80$  over 74 notes; both order vanilla < best.
- **Levels, no.** Judge 2 scores every note 2–4 points higher; plan recall means differ by 5 points ( $\rho = 0.54$ ).
- **Attribution, not at all.** On the same 74 notes, Qwen flags 4, DeepSeek 1, Gemma 4 flags 13; pairwise  $\kappa$  –0.02 (Qwen–DeepSeek), +0.29 (Qwen–Gemma), +0.12 (DeepSeek–Gemma); four notes are flagged by two judges and none by all three. Gemma also reverses the direction of the pipeline difference (vanilla 0.135 vs best 0.216 per note, against 0.08 vs 0.03 under Qwen). The misattribution penalty carries 40 points per event in the composite, so this is the term most exposed to judge choice, and the one dimension where the paper can make no directional claim at all.
- **Self-judging.** In Study 2 the writer and the judge are the same model; it cannot detect its own over-escalation (the anaphylaxis-vs-antihistamine disagreement is judged “contradicts” both ways).
- **Objective design.** Two failures were caused by the metric, not the model: citations removed (3.4) and unspoken plan items credited (3.5). Both were found by re-scoring, not by a new model.

- **RL sign disagreement** (3.7).
- Implication: use LLM judges to order pipelines and to audit objectives; calibrate at least the attribution judge against clinicians before any absolute claim.

## 0.8 6. Limitations

Mock consultations, UK primary care, 57 recordings; terse reference notes make absolute overlap scores low by construction; no chart context; judges uncalibrated; the vanilla pipeline is an unoptimized open baseline, not any vendor’s product; DeepSeek is both writer and judge 2 for the best configuration; the live view is a replay, and consult modality is inferred; the plan-recall audit and the live scoring use the same judge that generated the outputs.

## 0.9 6a. Data licensing and ethics

PriMock57 is released under CC BY 4.0 (repository LICENSE.md); ACI-Bench under CC BY 4.0 with a citation requirement (Yim et al. 2023); MTS-Dialog under CC BY 4.0; MedSynth’s repository and dataset card state no license as of 13 Sep 2026; the harness uses only its term strings (facts, not creative expression) to build the lexicon, and we will ask the authors for a licence statement before release. The lexicon derived from MTS-Dialog and MedSynth is released under CC BY 4.0 with attribution to both. Guideline pages fetched for the live view come from nhs.uk, whose content is released under the Open Government Licence v3 with the attribution “Contains information from NHS England, licensed under the current version of the Open Government Licence”; patient.info professional articles were read at run time and are not redistributed; Wikipedia extracts are CC BY-SA and are used only as model context. All consultations are role-played by clinicians and actors; no patient data, no protected health information, no clinical use. No IRB review was required.

## 0.10 7. Reproducibility

Code, lexicon, splits, prompts, run notebooks and cached model outputs are in the repository; every table’s source file is listed in the repository. All models are open weights; the two workstation GPUs and about \$49 of cloud time suffice to rerun everything except the 550B base-model run.

## 0.11 Figures and tables (list)

1. Table 1: ASR matrix (WER, term miss with CIs, DER, speakers) — from decoder-finding.
2. Figure 1: term miss vs WER scatter, decoder family coloured — drawn, below.
3. Table 2: held-out note metrics for vanilla / v1 / best / cited with bootstrap CIs — from runs/paper\_stats.md.
4. Figure 2: the search’s dev-vs-test trajectory — drawn, below.
5. Table 3: live-view versions with CIs — from runs/paper\_stats.md.

6. Figure 3: per-visit red flags and contradictions, v2 vs v5 — drawn, below.
7. Table 4: three-judge agreement. 8. Table 5: ACI-Bench fine-tuning with per-encounter CIs.
8. Appendix A: run log (the run log, the search log); Appendix B: prompts; Appendix C: verifier benchmark (Appendix C); Appendix D: fairness (Appendix D).

*Eight configurations of seven systems on the 57 PriMock57 recordings. WER spans about two points; medical-term loss spans about seven. Colour is decoder family.*

*Grey line: dev-set composite (20 visits) for each of 71 logged iterations. Dots: held-out test composite (37 visits) for the 14 iterations that earned a test run; green = accepted, red = a dev gain that failed the gate. Five acceptances, all before iteration 50.*

*Bars: unique red flags raised per visit (amber = v2, green = v5). Crosses: contradicting plan suggestions at the end of the visit (top row v2, bottom row v5). v5 raises flags in two visits, both genuine.*

## **0.12 To do before submission**

- Done 13 Sep: related work, Figures 1-3, per-encounter CIs for the fine-tuning table, a third attribution judge (result: no agreement, so the attribution claims are now stated as unresolved rather than as a two-thirds reduction).
- Related work: drafted and verified (section 2a), all twelve author lists complete.
- A clinician read of 20 notes and the six live-view items as a “v2 paper” hook, or as a small appendix if one can be arranged before the deadline.

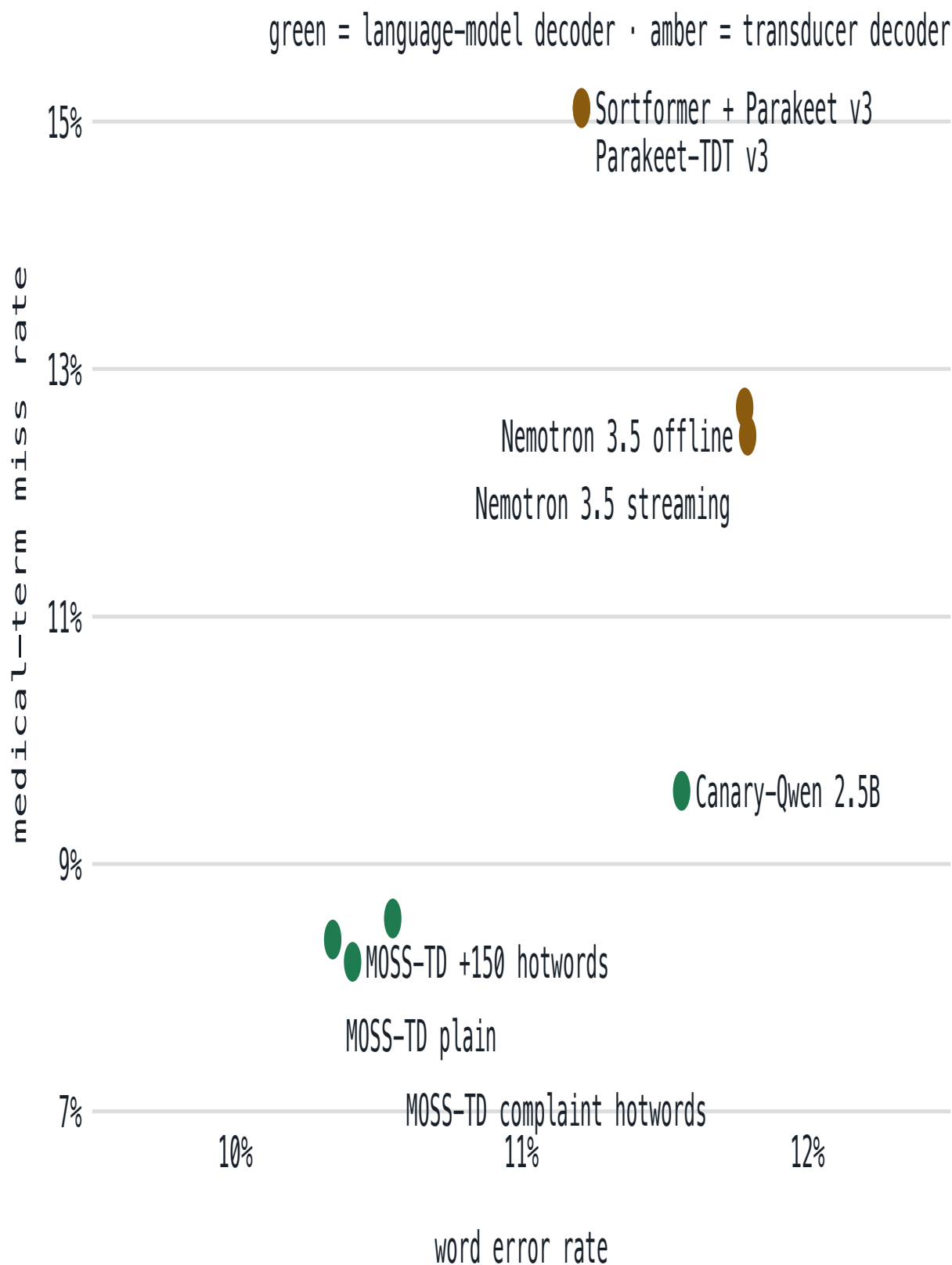


Figure 1: Medical-term miss rate against word error rate, seven open ASR systems

52  
 grey: dev composite (20 visits) · dots: held-out test (37) · green = accepted, red = dev gain that failed test

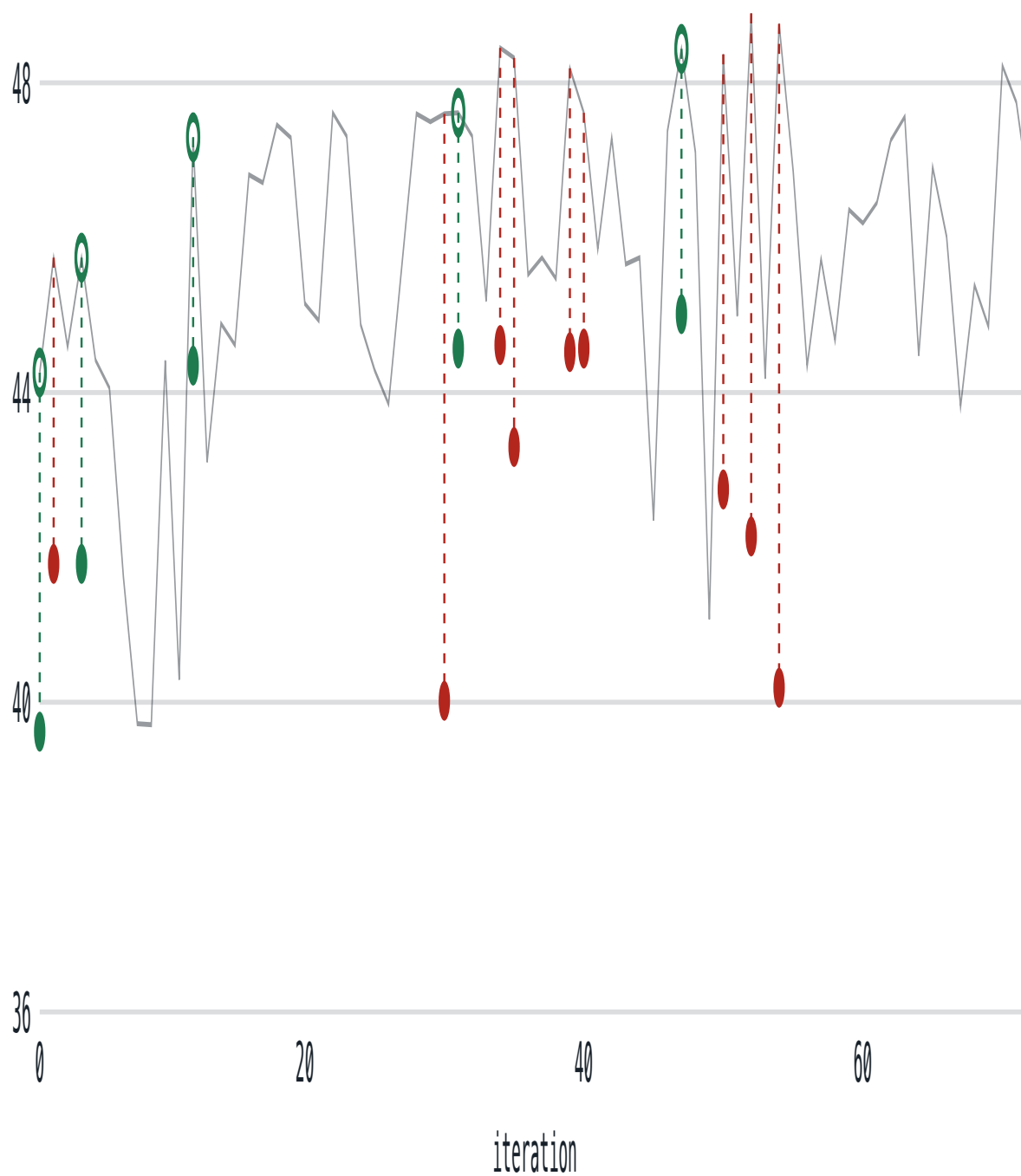


Figure 2: Overnight search: dev composite per iteration, held-out test where run

